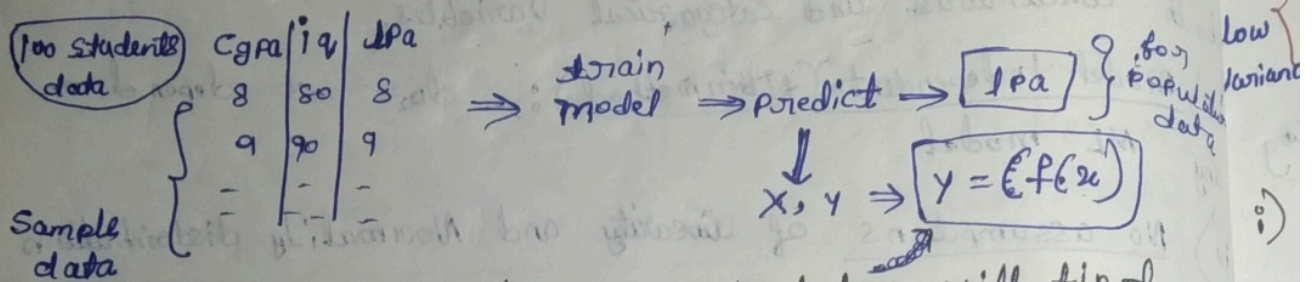


Bias Variance Tradeoff

→ Bias → Overfitting → tradeoff
 → Variance → Underfitting → Bias Variance decomposition



So we train model on sample data, will find coefficients and Equation. But we use that same Equation for population data.

$$y = f(x) + \text{error}$$

ml model irreducible error

$$y = f(x)$$

population data

↓
close estimation of y

$$\hat{y} = f'(x)$$

$$y - \hat{y} \Rightarrow f(x) - f'(x)$$

reducible error

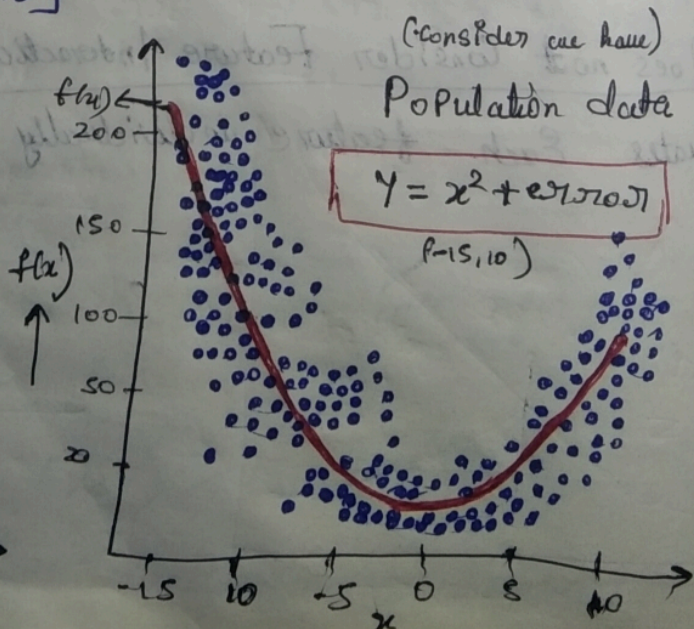
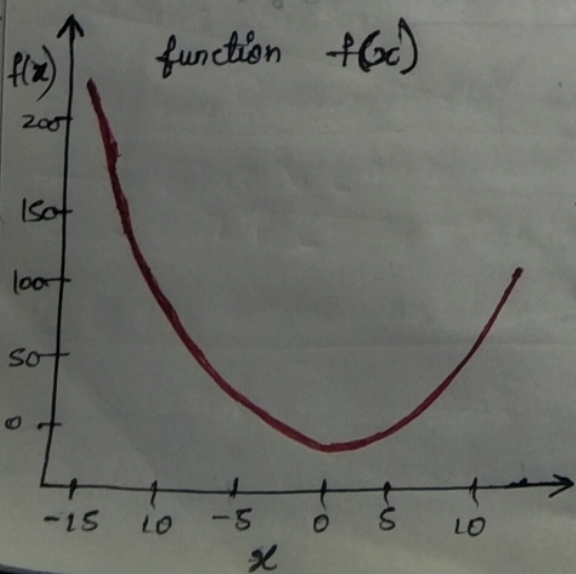
when the data size will increase on applying ml technique we can reduce the error & can improve the model.

Bias Variance Tradeoff ⇒ Reduceable Error

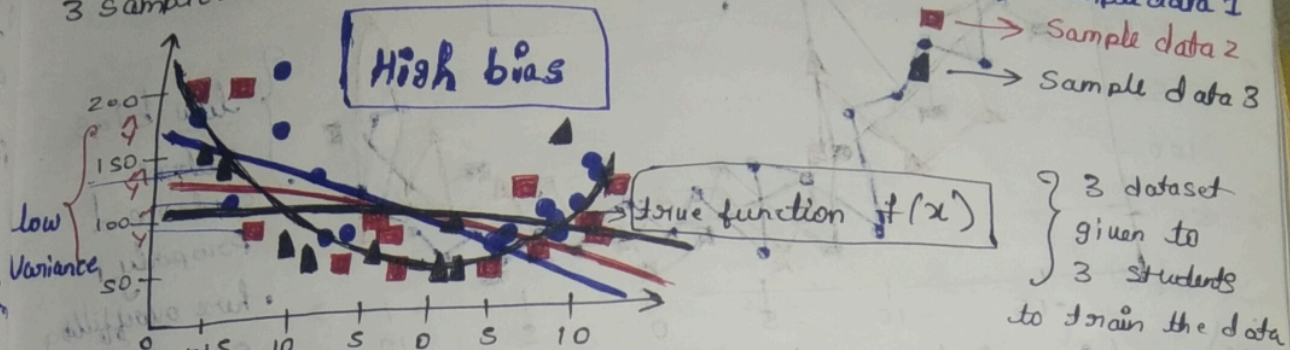
$$\text{Reduceable Error} = \text{Bias}^2 + \text{Variance}$$

Bias Variance TradeOff :-

$$y = f(x) = x^2 \quad [-15, 10]$$



→ from the Population dataset, we are taking 3 samples. and plotting on the graph.



i) applying linear model on the dataset

Bias $\Rightarrow$ Inability of a model to fit the training data.

The goal of ML model is:

→ Learn Patterns from training data.

→ Make good prediction on new (unseen) data.

If the model performs well only on training data but fails on new data $\rightarrow$ not useful

• here the bias and Variance come in

i) Bias:- Bias is the error caused by overly simple assumptions made by the model.

→ model is too simple

→ It ignores important features Patterns

→ It assumes data is simple than it is

→ model cannot learn true relationship

→ Predictions are consistently wrong

Characters:-

→ Training error $\Rightarrow$ high

→ Testing error $\Rightarrow$ high

model performs poor

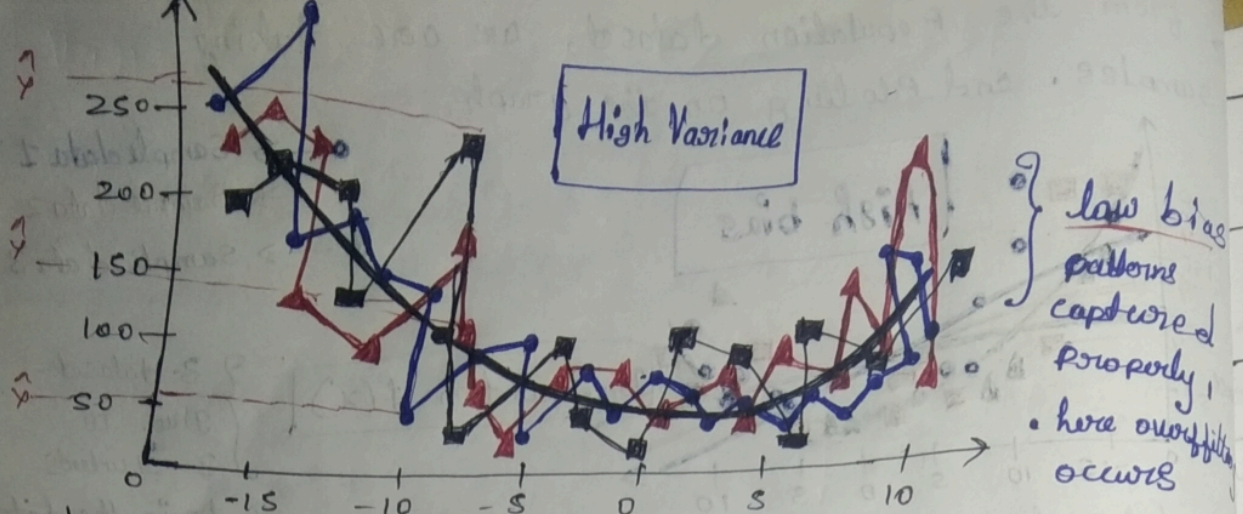
"In simple regression lines are far from the data points."

Examples of High bias model:

• Simple Linear Regression for complex data

• Shallow decision tree

• Very strong regularization.



ii) applying Polynomial regression

ii) Variance: Variance is the error caused by the model being too sensitive to the training data

- model is too complex & learns noise + patterns
- model fits the data extremely well
- But fails on unseen data.

result: - Overfitting.

Examples of high Variance models:

- Very deep decision trees
- High degree Polynomial regression
- Neural Networks with little data.

Bias Variance TradeOff:

Bias Variance Tradeoff is the balance between underfitting and (high bias) and overfitting (high variance) to achieve the lowest possible prediction error on unseen data.

→ You cannot reduce both bias and Variance at the same time. This is the trade off.

i) if you increase model complexity

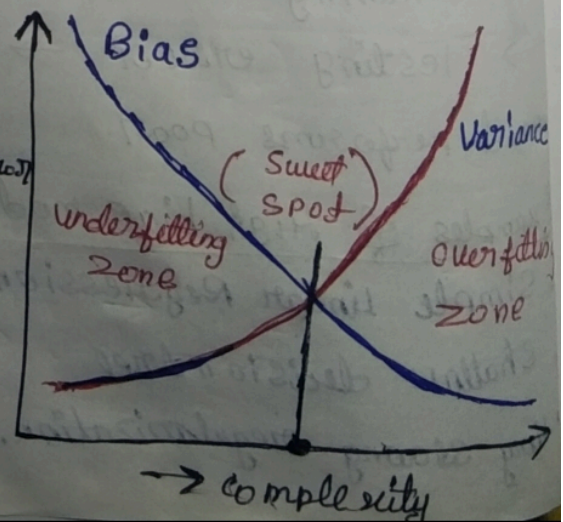
→ Bias ↓ (model learns more

→ Var ↑ (model learns noise + pattern)

ii) If you decrease model complexity

→ Var ↓ (model stable)

→ Bias ↑ (model too simple)



- 1) How would you define bias and Variance mathematically?
- 2) How is bias and Variance related to overfitting and underfitting mathematically.
- 3) Why there is a tradeoff between bias & Variance mathematically.

Expected Value $\approx$

Population mean

Expected Value and Variance:-

Expected Value represents average outcome of a random Variable over large number of trial or Experiments

Ex: Tossing a dice 1 Jack times, what is the mean = ?

X = Tossing dice

$$E[X] = X_1 P(X_1) + X_2 P(X_2) + \dots + X_n P(X_n)$$

$$\Rightarrow 1 \times \left(\frac{1}{6}\right) + 2 \times \left(\frac{1}{6}\right) + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$$

$$\Rightarrow \frac{1+2+3+4+5+6}{6} = \frac{21}{6} \Rightarrow 3.5 \quad \text{over all avg / mean}$$

Ex 2:- 5 3 4 5 3 5 $\Rightarrow$ mean = ?

$$\Rightarrow \frac{5+3+4+5+3+5}{6}$$

$$\Rightarrow \frac{25}{6}$$

$$\Rightarrow \frac{3(5) + 2(3) + 1(4)}{6}$$

$$\Rightarrow \frac{3}{6}(5) + \frac{2}{6}(3) + \frac{1}{6}(4)$$

$$E[X] \Rightarrow X_1 P(X_1) + X_2 P(X_2) + X_3 P(X_3)$$

i) discrete random variable (X)

$$E[X] = \sum_{i=1}^n X_i P(X_i)$$

ii) Continuous random variable

$$E[X] = \int X_i f(X_i) dx$$

Variance :- $\text{Var}[X] = \text{Variance of population}$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

aspect

measures

meaning

symbol

Expected Value

$\rightarrow$ central value

$\rightarrow$ average / mean

$E[X]$

Variance

$\rightarrow$ spread

$\rightarrow$ Dispersion

$\text{var}[X]$

$$\text{Var} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \Rightarrow E[(X - E[X])^2] \Rightarrow \text{Var}[X]$$

$\frac{\text{Sum}}{n} = \text{avg} / \text{mean} = E[\]$, $\bar{X} \Rightarrow \text{avg} = E[X]$

- X is a random variable
- X represents outcome of a random experiment

$$\begin{aligned} \text{Var}[X] &= E[(X - E[X])^2] \\ &\Rightarrow E[X^2 + (E[X])^2 - 2XE[X]] \Rightarrow E[X^2] + E[(E[X])^2] - E[2XE[X]] \\ &\Rightarrow E[X^2] + E[(E[X])^2] - E[2]E[X]E[E[X]] \Rightarrow E[X^2] + E[(E[X])^2] - 2E[X]E[X] \\ &\Rightarrow E[X^2] + (E[X])^2 - 2(E[X])^2 \\ &\Rightarrow E[X^2] - (E[X])^2 \end{aligned}$$

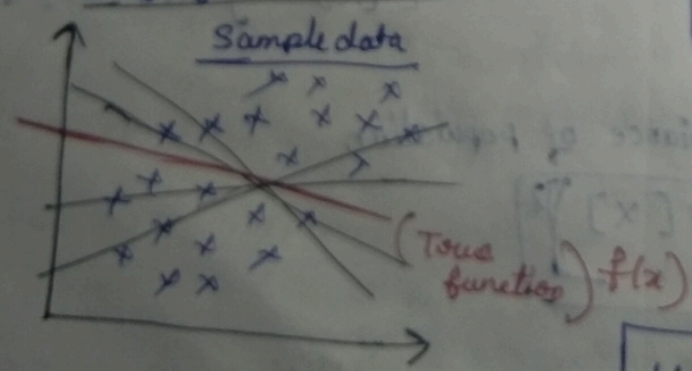
$E[X+1] = E[X] + 1$
 $E[XY] = E[X]E[Y]$ give $(X) \perp (Y)$ are independent
 $E[\text{constant}] = \text{constant}$
 $E[X] = \text{number}$

$\text{Var}[X] = E[X^2] - (E[X])^2$ discrete continuous

What Exactly are Bias and Variance Mathematically?

i) Bias:-

- Bias refers to the systematic error that a model introduce because it cannot capture the relationship in data
- It represents the difference b/w the Expected prediction of our model and correct value which we are trying to predict.



$$\text{True function} \Rightarrow f(x)$$

$$\text{trained ML model function} \Rightarrow f'(x)$$

$$\text{bias}(f'(x)) = E[f'(x)] - f(x)$$

Think like this:-

- model $\Rightarrow f'(x) \rightarrow$ guess
- $f(x) \rightarrow$ gods function (unknown)
- Data $\rightarrow$ samples from this function

we can't compare directly to gods func, so we judge by Performance Data

$$\text{Bias}(f'(x)) = E[f'(x)] - f(x)$$

→ Let's decode each of them

what is $f(x)$:-

- $f(x)$ is a true (real) function
- It is the actual relationship b/w input and output in the real world.
- ~~we~~ we never know $f(x)$ in real life

what is $f'(x)$:-

- This our model
- Learned from data (Linear Regression, tree, NN etc)
- It is an estimate of the true function
guessing

what is $E[f'(x)]$:-

- This is Expected prediction of the model.
- Train the same model (Ex:- Linear Regression)
- On many different sample dataset (from the same population data)
- Take average prediction at point x

This average = $E[f'(x)]$

Bias measures :- How far is the average ~~prediction~~ model prediction from the true function

$$\text{Var}(f'(x)) = E \left[\left(f'(x) - E[f'(x)] \right)^2 \right]$$

what is $f'(x)$:-

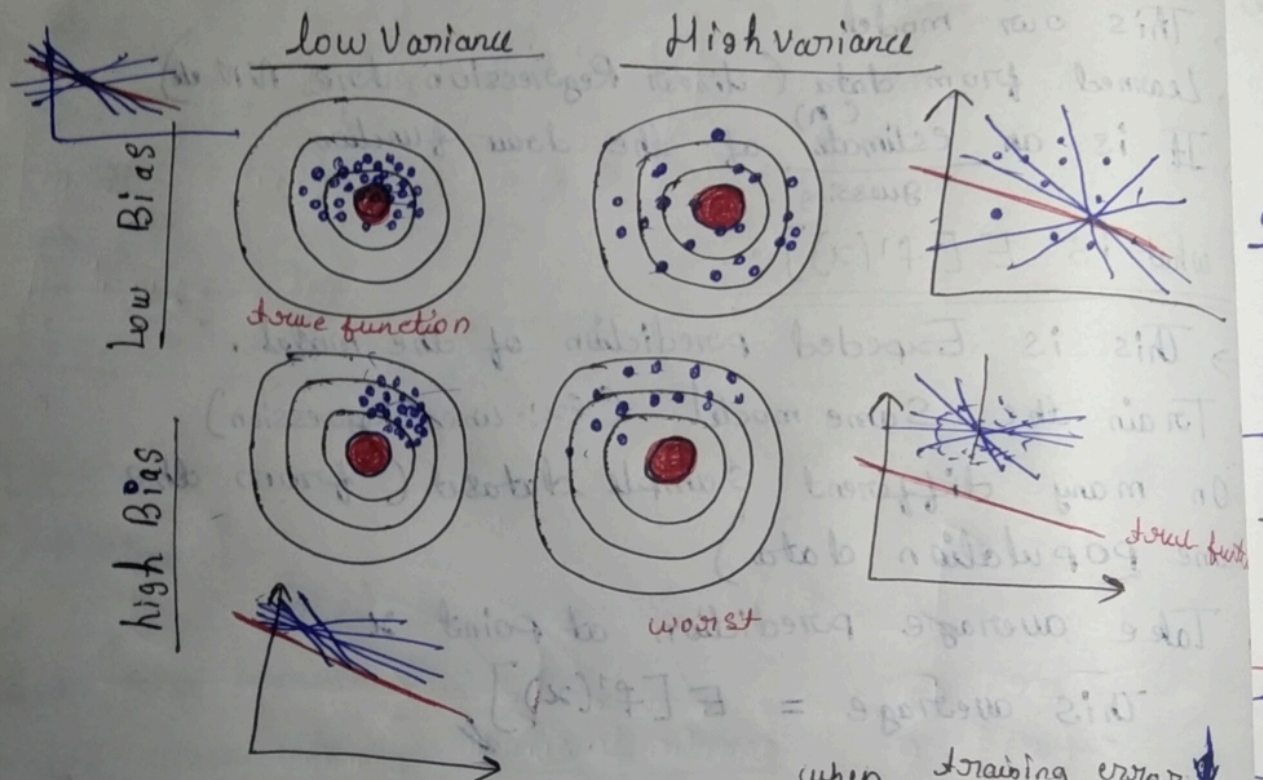
- The model Prediction at point x
- Learned from a training dataset
- Changes if the training data changes

what is $E[f'(x)]$

- The average prediction
- over many different training datasets (refer high bias & high var diagram)
- using the same model type.

Variance measures how much the model's predictions change when the training data changes.

- if predictions change a lot → High Variance
- Predictions are stable ⇒ Low Variance



how to know bias is reducing

- not by computing formula
- But by observing: → training error ↓
→ validation error ↓
if both reduce bias will ↓

How to reduce bias

- increase model complexity,
- add better features → feature Engineer domain knowledge
- Reduce regularization
- Change model family
linear → Tree-based → Neural networks

when training error ↓
testing error ↑

How to know Variance reducing

- Training Error slightly increases
- test error decreases
- Gap b/w train & test reducing

how to reduce Variance

- using more train data
- Reduce model complexity
- use regularization
- Ensemble methods
- Cross validation

Bias Variance Decomposition:-

when a model makes a error, we want to know where the error is coming from?

→ Bias Variance Decomposition is a way of analyzing a learning algorithm's generalization error with respect to a particular problem

→ Bias Variance Decomposition breaks the expected prediction error into three parts, so we can reason about the model behavior.

$$\text{Loss} = \text{bias} + \text{Variance} + \text{irreducible Error}$$

$$\text{Loss} = \underbrace{[\text{bias}^2 + \text{Variance}]}_{\text{reducible Error}} + \underbrace{\text{Var}[\epsilon]}_{\text{noise}}$$

Age	iq	Age	Pred
-	-	8	9.2
-	-	8	8.1
-	-	7	6.3
-	-	9	10.1

Linear Regression

reducible error

irreducible Error

MSE

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

→ minimize MSE

Consider:-

$$\text{irreducible Error} \begin{cases} \rightarrow \text{mean} = 0 \\ \rightarrow \text{Var} = \sigma^2 \end{cases}$$

Derivation:- $\text{MSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \Rightarrow E[(y - \hat{y})^2]$

$$\begin{aligned} y &= f(x) + \epsilon \Rightarrow \theta + \epsilon \\ \hat{y} &= \hat{f}(x) \Rightarrow \hat{\theta} \end{aligned}$$

$$\Rightarrow E[(\theta + \epsilon - \hat{\theta})^2]$$

$$\Rightarrow E[(\underbrace{\theta - \hat{\theta}}_a + \epsilon)^2]$$

$$\Rightarrow E[\underbrace{(\theta - \hat{\theta})^2}_{a^2} + \underbrace{\epsilon^2}_{b^2} + 2\epsilon(\theta - \hat{\theta})]$$

$$\Rightarrow E[(\theta - \hat{\theta})^2] + E[\epsilon^2] + E[2\epsilon(\theta - \hat{\theta})]$$

$$\Rightarrow E[(\theta - \hat{\theta})^2] + E[\epsilon^2] + E[2]E[\epsilon]E[(\theta - \hat{\theta})]$$

$$E[\epsilon^2] \Rightarrow \begin{cases} \text{Var}(\epsilon) \Rightarrow \sigma^2 = E[(\epsilon - E[\epsilon])^2] \\ \rightarrow E[\epsilon] = 0 \end{cases} \text{mean of irreducible}$$

$$= E[(\epsilon - 0)^2]$$

$$\Rightarrow E[\epsilon^2]$$

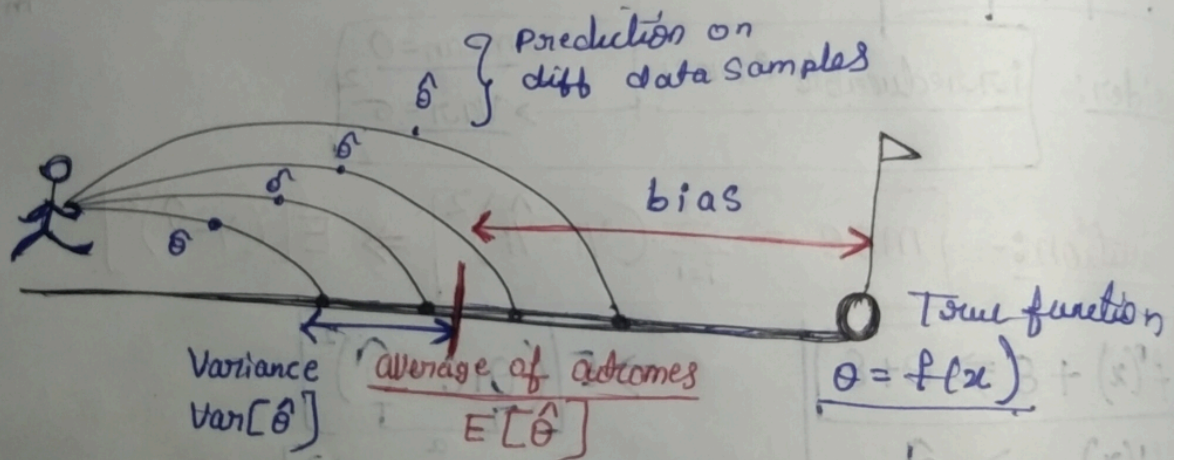
$$mse = \underbrace{E[(\theta - \hat{\theta})^2]} + \underbrace{var(E)}_{\text{irreducible Error}}$$

$$\begin{aligned} E[(\theta - \hat{\theta})^2] &\Rightarrow E\left[\underbrace{(\theta - E[\hat{\theta}])}_a + \underbrace{(E[\hat{\theta}] - \hat{\theta})}_b\right]^2 \\ &= E\left[(\theta - E[\hat{\theta}])^2 + (E[\hat{\theta}] - \hat{\theta})^2 + 2(\theta - E[\hat{\theta}])(E[\hat{\theta}] - \hat{\theta})\right] \\ &= E[(\theta - E[\hat{\theta}])^2] + E[(E[\hat{\theta}] - \hat{\theta})^2] + E[2(\theta - E[\hat{\theta}])(E[\hat{\theta}] - \hat{\theta})] \\ &\quad \underbrace{(\theta - E[\hat{\theta}])^2}_{\text{Bias}^2} + \underbrace{(E[\hat{\theta}] - \hat{\theta})^2}_{\text{Var}} + \underbrace{E[2(\theta - E[\hat{\theta}])(E[\hat{\theta}] - \hat{\theta})]}_{\text{Covariance term}} \end{aligned}$$

$$\boxed{mse = \underbrace{\text{reducible}} + \underbrace{\text{irreducible}}}$$

$$\boxed{mse = \underbrace{(\text{bias})^2}_{\text{Bias}} + \underbrace{\text{Variance}} + \underbrace{var(E)}_{\text{Variance}}}$$

Analogy :- golfer



why $(\text{bias})^2 \Rightarrow$ due to loss function we considered

$$mse = (y_i - \hat{y})^2$$

$\Rightarrow$ to handle negative values

while training
loss (in ML models)

loss (in ML models) branches into:
bias, var, Target (noise) in

i) underfitting $\rightarrow$ use different ML algorithms

ii) overfitting $\rightarrow$ regularization